

Empirical Bayes: Part I (Step Count Problem)

Statistical Reinforcement Learning Lab

Problem Statement

In our study, we have a population of 30 unique individuals and are interested in investigating the number of steps each person takes in a 30-minute interval. We designate the true mean steps for each of 30 individuals as $\mu_1, \mu_2, \dots, \mu_i$ for $i = 1, 2, \dots, 30$ ($n = 30$).

During the study, we use a device to measure the number of steps each individual takes during a 30-minute interval. We use the device to perform this measurement once for each individual.

Now, we have two data points for each individual:

1. The device's step count for that individual.
2. A measure of uncertainty provided by the device, which describes how sure the device is that it counted the number of steps correctly.

We designate the observed step counts as $x_1, x_2, \dots, x_i$ and the measures of uncertainty as $\sigma_1, \sigma_2, \dots, \sigma_i$. Thus, our data for each of the 30 individuals consists of one pair of values, (x_i, σ_i) .

The true means and hyperparameters are unknown to us. The observed step counts and measures of uncertainty are our data and thus are known to us. How can we estimate μ_0 and τ^2 using this data?

Empirical Bayes Solution

We will approach this problem using Raudenbush & Bryk's empirical Bayes model for meta-analyses [1].

First, we assume that each person's true mean steps is a draw from the same underlying normal distribution with some non-zero variance τ^2 :

$$\mu_i \sim \mathcal{N}(\mu_0, \tau^2).$$

We also assume each person's step count on a given day is a draw from a normal distribution centered at that person's mean, μ_i , with non-zero variance σ_i^2 . (We assume that the device's uncertainty measurement, σ_i^2 , accurately captures the true within-person variance.)

Then, for this problem, τ^2 designates population variance while σ_i^2 designates within-person variance.

By equation 8 (p. 78) and equation 18 (p. 80) [1], we can model each individual's observation is a draw from a normal distribution centered at that person's mean μ_i . The variance of this observation is the sum of within-person and population variance. Therefore, the distribution of the observed data $\mathbf{x}$ is:

$$x_i \sim \mathcal{N}(\mu_0, \sigma_i^2 + \tau^2), \quad x_i \mid \mu_i \sim \mathcal{N}(\mu_i, \sigma_i^2).$$

Let us note that our model parallels Raudenbush & Bryk's model as follows [1]: $d_i = x_i$, $W'_i = 1$, $\gamma = \mu_0$, $v_i = \sigma_i^2$. Recall that in our problem, σ_i^2 is known to us as it is provided by the device, but τ^2 and μ_0 are both unknown. Thus, we are in the scenario with unknown variance and will reference the *Variance estimation* section (pp. 80-82).

We can derive the likelihood and log-likelihood functions of τ^2 and μ_0 from the normal PDF as follows:

$$L(\mu_0, \tau^2) = \prod_{i=1}^n \mathcal{N}(x_i \mid \mu_0, \sigma_i^2 + \tau^2) \quad (1)$$

$$L(\mu_0, \tau^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi(\sigma_i^2 + \tau^2)}} \exp\left[-\frac{(x_i - \mu_0)^2}{2(\sigma_i^2 + \tau^2)}\right] \quad (2)$$

$$\ell(\mu_0, \tau^2) = -\frac{1}{2} \sum_{i=1}^n \left\{ \log(\sigma_i^2 + \tau^2) + \frac{(x_i - \mu_0)^2}{\sigma_i^2 + \tau^2} \right\} \quad (3)$$

We can rewrite the conditional log-likelihood for μ_0 as follows:

$$Q(\mu_0; \tau^2) = \sum_{i=1}^n w_i(\tau^2) (x_i - \mu_0)^2, \quad w_i(\tau^2) = \frac{1}{\sigma_i^2 + \tau^2} \quad (4)$$

Next, we differentiate w.r.t. μ_0 to determine the MLE estimator for μ_0 conditional on τ^2 . Here, we are considering τ^2 as a constant and hence the $\log(\sigma_i^2 + \tau^2)$ term is dropped.

$$\frac{\partial Q}{\partial \mu_0} = -2 \sum_{i=1}^n w_i(\tau^2) (x_i - \mu_0) = 0 \quad (5)$$

$$\mu_0^*(\tau^2) = \frac{\sum_{i=1}^n w_i(\tau^2) x_i}{\sum_{i=1}^n w_i(\tau^2)} \quad (6)$$

We have effectively written the conditional MLE of μ_0 in terms of τ^2 , so we only have one unknown variable remaining (τ^2). We will replace μ_0 with our conditional MLE in the original log-likelihood equation. Now, we have:

$$\ell_p(\tau^2) = -\frac{1}{2} \left\{ \sum_{i=1}^n \log(\sigma_i^2 + \tau^2) + Q_1(\tau^2) \right\}, \quad Q_1(\tau^2) = \sum_{i=1}^n w_i(\tau^2) (x_i - \mu_0^*(\tau^2))^2 \quad (7)$$

We define the MLE of τ^2 as:

$$\hat{\tau}^2 = \arg \max_{\tau^2 \geq 0} \ell_p(\tau^2) \quad (8)$$

Now, we can substitute this estimate back into our conditional MLE estimator for μ_0 to obtain an estimate for μ_0 :

$$\hat{\mu}_0 = \mu_0^*(\hat{\tau}^2) = \frac{\sum_{i=1}^n (\sigma_i^2 + \hat{\tau}^2)^{-1} x_i}{\sum_{i=1}^n (\sigma_i^2 + \hat{\tau}^2)^{-1}} \quad (9)$$

Now, we can write the joint likelihood function for $\mu_i, \{x_1, \dots, x_n\}$ as follows:

$$L(\mu_i; \mathbf{x}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left\{-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right\} \cdot \frac{1}{\sqrt{2\pi\tau^2}} \exp\left\{-\frac{(\mu_i - \mu_0)^2}{2\tau^2}\right\} \quad (10)$$

$$\ell(\mu_i; \mathbf{x}) = -\frac{1}{2} \sum_{i=1}^n \left[\log(2\pi\sigma_i^2) + \frac{(x_i - \mu_i)^2}{\sigma_i^2} + \log(2\pi\tau^2) + \frac{(\mu_i - \mu_0)^2}{\tau^2} \right] \quad (11)$$

Pattern-Matching Approach

First, we take the partial derivative w.r.t. μ_i to derive the posterior mean, substituting in the MLE estimators we derived above.

$$\frac{\partial \ell(\mu_i; \mathbf{x})}{\partial \mu_i} = \frac{x_i - \mu_i}{\sigma_i^2} + \frac{\mu_0 - \mu_i}{\tau^2} \quad (12)$$

$$\Rightarrow \mu_i \left(\frac{1}{\sigma_i^2} + \frac{1}{\hat{\tau}^2} \right) = \frac{x_i}{\sigma_i^2} + \frac{\hat{\mu}_0}{\hat{\tau}^2} \quad (13)$$

$$\Rightarrow \mu_i = \lambda_i x_i + (1 - \lambda_i) \hat{\mu}_0, \quad \lambda_i = \frac{\hat{\tau}^2}{\sigma_i^2 + \hat{\tau}^2} \quad (14)$$

Second, we derive the MLE of the posterior variance by Fisher Information:

$$\frac{\partial^2 \ell}{\partial \mu_i^2} = - \left(\frac{1}{\sigma_i^2} + \frac{1}{\tau^2} \right) \quad (15)$$

$$\Rightarrow \text{Var}(\mu_i | x_i) = \left(- \frac{\partial^2 \ell}{\partial \mu_i^2} \right)^{-1} = \left(\frac{1}{\sigma_i^2} + \frac{1}{\tau^2} \right)^{-1} \quad (16)$$

To conclude, we have:

$$\text{Prior: } \mu_i \sim \mathcal{N}(\hat{\mu}_0, \tau^2) \quad (17)$$

$$\text{Posterior: } \mu_i | \mathbf{x} \sim \mathcal{N} \left(\lambda_i x_i + (1 - \lambda_i) \hat{\mu}_0, \left(\frac{1}{\sigma_i^2} + \frac{1}{\hat{\tau}^2} \right)^{-1} \right), \quad \lambda_i = \frac{\sigma_i^2}{\sigma_i^2 + \hat{\tau}^2}. \quad (18)$$

Posterior Distribution Derivation

Recall that we defined above:

$$x_i \sim \mathcal{N}(\mu_0, \sigma_i^2 + \tau^2), \quad x_i | \mu_i \sim \mathcal{N}(\mu_i, \sigma_i^2).$$

We will use Bayes' rule to derive the full posterior distribution:

$$p(\mu_i | x_i) = \frac{p(x_i | \mu_i) p(\mu_i)}{p(x_i)} \propto p(x_i | \mu_i) p(\mu_i) \quad (19)$$

We can write out the probability distribution functions based on the distributions defined above:

$$p(\mu_i) = \frac{1}{\sqrt{2\pi\tau^2}} \exp \left[-\frac{(\mu_i - \mu_0)^2}{2\tau^2} \right], \quad p(x_i | \mu_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left[-\frac{(x_i - \mu_i)^2}{2\sigma_i^2} \right] \quad (20)$$

We can multiply these two functions, dropping constants and combining the exponents.

$$p(\mu_i | x_i) \propto \exp \left[-\frac{(x_i - \mu_i)^2}{2\sigma_i^2} - \frac{(\mu_i - \mu_0)^2}{2\tau^2} \right] \quad (21)$$

$$= \exp \left[-\frac{\tau^2(x_i - \mu_i)^2 + \sigma_i^2(\mu_i - \mu_0)^2}{2\sigma_i^2\tau^2} \right] \quad (22)$$

$$= \exp \left[-\frac{\tau^2(x_i^2 - 2x_i\mu_i + \mu_i^2) + \sigma_i^2(\mu_i^2 - 2\mu_0\mu_i + \mu_0^2)}{2\sigma_i^2\tau^2} \right] \quad (23)$$

$$= \exp \left[-\frac{(\tau^2 + \sigma_i^2)\mu_i^2 - 2(\tau^2x_i + \sigma_i^2\mu_0)\mu_i + (\tau^2x_i^2 + \sigma_i^2\mu_0^2)}{2\sigma_i^2\tau^2} \right] \quad (24)$$

$$= \exp \left[-(\tau^2 + \sigma_i^2) \left(\mu_i^2 - 2 \cdot \frac{\tau^2x_i + \sigma_i^2\mu_0}{\tau^2 + \sigma_i^2} \mu_i \right) \right] \quad (25)$$

$$= \exp \left[-(\tau^2 + \sigma_i^2) \left(\left(\mu_i - \frac{\tau^2x_i + \sigma_i^2\mu_0}{\tau^2 + \sigma_i^2} \right)^2 - \left(\frac{\tau^2x_i + \sigma_i^2\mu_0}{\tau^2 + \sigma_i^2} \right)^2 \right) \right] \quad (26)$$

$$= \exp \left[-\frac{\tau^2 + \sigma_i^2}{2\sigma_i^2\tau^2} \left(\mu_i - \frac{\tau^2x_i + \sigma_i^2\mu_0}{\tau^2 + \sigma_i^2} \right)^2 \right] \quad (27)$$

By the Normal-Normal Conjugacy, we can pattern match this exponent to the exponent of the normal PDF, $\exp\left(-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right)$, to derive the posterior distribution.

$$\mu_i | x_i \sim \mathcal{N} \left(\frac{\tau^2x_i + \sigma_i^2\mu_0}{\tau^2 + \sigma_i^2}, \frac{1}{\frac{\tau^2 + \sigma_i^2}{\sigma_i^2\tau^2}} \right) \quad (28)$$

$$\mu_i | x_i \sim \mathcal{N} \left(\frac{\tau^2}{\sigma_i^2 + \tau^2} x_i + \frac{\sigma_i^2}{\sigma_i^2 + \tau^2} \mu_0, \frac{\sigma_i^2\tau^2}{\sigma_i^2 + \tau^2} \right) \quad (29)$$

This yields the same result as display (18) above. *However*, we should be aware that this pattern-matching approach with the Normal-Normal Conjugacy only holds in select scenarios with strict assumptions. The full Bayesian derivation, which involves a Hessian matrix, should be computed for most scenarios.

References

- [1] Stephen W. Raudenbush and Anthony S. Bryk. Empirical bayes meta-analysis. *Journal of Educational Statistics*, 10(2):75–98, 1985.

Empirical Bayes: Part II (Two-Armed Bandit Simulation Outline)

Statistical Reinforcement Learning Lab

This is a two-armed bandit simulation. The basic notation is as follows:

- Participants (i): There are 50 participants denoted as $i = 1, i = 2, \dots, i = 50$.
- Decision times (t): There are 11 decision times denoted as $t = 0, t = 1, t = 2, \dots, t = 11$.
- Actions (A): At every time point, the participant chooses between two possible actions denoted as $a = 0$ and $a = 1$ respectively.
- Reward (R): Based on the action the participant selects, the participant will receive a reward. r may be equal to any real number.

I. Defining the underlying environment

Every time a participant chooses an action, the underlying environment will generate a reward either completely at random (e.g., using a random number generator) or performing draws from distributions that may or may not be Normal.

II. Learning algorithm

The learning algorithm will observe all actions and rewards up to and including time $t = 11$ on individual i . Based on some method, the learning algorithm will estimate the value of a parameter called $\theta_{i,a}$ for each individual and action. These estimates will be called $x_{i,a,t}$, $i = 1, \dots, 50$; $t = 11$; $a = 0, 1$.

The learning algorithm will also output a measure of its uncertainty in the estimate $x_{i,a,t}$ of the parameter $\theta_{i,a}$ for each individual and action. This uncertainty is denoted by $\sigma_{i,a,t}^2$.

After each time point t , the learning algorithm will share updated $x_{i,a,t}, \sigma_{i,a,t}^2$ values with the action selection strategy. As there are 50 participants and 2 possible actions, the action selection strategy will receive a total of 200 data points at every time point.

III. Action Selection Strategy

The action selection strategy believes that the $x_{i,a,t}$ produced by the learning algorithm are estimates of the true mean (of some parameter of interest that determines the reward) of each individual-action combination at time t . It also believes that $\sigma_{i,a,t}^2$ is the learning algorithm's measure of uncertainty regarding its guess at the mean.

Specifically, the action selection strategy is built on the following premise (which may or may not align with how the data was actually generated):

(1) *Distribution of the learning algorithm's estimates of the mean ($X_{i,a,t}$)*

$\theta_{i,a}$ is the mean of some parameter that determines the reward distribution.

$$X_{i,a,t} \sim N(\theta_{i,a}, \sigma_{i,a}^2)$$

The action selection strategy believes that the learning algorithm is attempting to guess $\theta_{i,a}$ and will produce a number of estimates at each time point ($x_{i,a,t}$), which it calls $X_{i,a,t}$ and

believes assumes the following Normal distribution. The learning algorithm also provides a measure of uncertainty for each estimate $\sigma_{i,a,t}^2$, which the action selection strategy believes to equal $v_{i,a}$ (the variance of the learning algorithm's estimates of the mean) in this model.

(2) *Distribution of the true means ($\theta_{i,a}$)*

Furthermore, the action selection strategy believes that the true means $\theta_{i,a}$ of each individual-action combination follow an underlying Normal distribution.

$$\begin{aligned}\theta_{i,a=0} &\sim N(\mu_{a=0}, \tau_{a=0}^2) \\ \theta_{i,a=1} &\sim N(\mu_{a=1}, \tau_{a=1}^2)\end{aligned}$$

Specifically, the action selection strategy believes that there is a population mean and a between-person variance represented by hyperparameters μ_a and τ_a^2 respectively.

Based on the action selection strategy's model of the data specified above, the action selection strategy will do the following after receiving the $x_{i,a,t}$ and $\sigma_{i,a,t}^2$ at each time point.

1. First, the action selection strategy will produce estimates for the population-level hyperparameters using the data. The action selection strategy will first estimate the variance and then the mean.

$$\hat{\tau}_a^2 = \arg \max_{\tau_a^2 \geq 0} \ell_p(\tau_a^2), \quad \hat{\mu}_a = \frac{\sum_{i=1}^n \left(\hat{\tau}_a^2 + \sigma_{i,a,t}^2 \right)^{-1} x_{i,a,t}}{\sum_{i=1}^n \left(\hat{\tau}_a^2 + \sigma_{i,a,t}^2 \right)^{-1}}.$$

2. Now, the learning algorithm will perform an update on the estimates of $\theta_{i,a}$ for each individual.

$$\text{Prior: } \theta_{i,a} \sim \mathcal{N}(\hat{\mu}_a, \hat{\tau}_a^2)$$

$$\text{Posterior: } \theta_{i,a} \mid x_{i,a,t}, \sigma_{i,a,t}^2 \sim \mathcal{N}\left(\lambda_{i,a,t} x_{i,a,t} + (1 - \lambda_{i,a,t}) \hat{\mu}_a, \left(\frac{1}{\sigma_{i,a,t}^2} + \frac{1}{\hat{\tau}_a^2} \right)^{-1}\right)$$

Aligned with empirical Bayes, we notice that there is a shrinkage term $\lambda_{i,a,t}$ that weighs the new information based on a comparison between its variance and our estimate of the population-level variance.

$$\lambda_{i,a,t} = \frac{\hat{\tau}_a^2}{\hat{\tau}_a^2 + \sigma_{i,a,t}^2}$$

Please see the "Step Count Problem" displays 19-30 for derivation.

3. Now, using the properties of the Normal distribution, we can determine the probability that a given arm will result in the greater value based on the two posterior distributions.

Recall that the action strategy believes that the rewards associated with the two arms are independent Normals, and the difference of two independent Normals is Normal. Thus, the action strategy believes:

$$\theta_{i,1} - \theta_{i,0} \sim \mathcal{N}\left(\left[\lambda_{i,1,t}x_{i,1,t} + (1 - \lambda_{i,1,t})\hat{\mu}_1\right] - \left[\lambda_{i,0,t}x_{i,0,t} + (1 - \lambda_{i,0,t})\hat{\mu}_0\right], \left(\sigma_{i,1,t}^{-2} + \hat{\tau}_1^{-2}\right)^{-1} + \left(\sigma_{i,0,t}^{-2} + \hat{\tau}_0^{-2}\right)^{-1}\right)$$

Thus, the probability of a random draw from one arm yielding a greater value is:

$$P(\theta_{i,1} > \theta_{i,0}) = \Phi\left(\frac{\left[\lambda_{i,1,t}x_{i,1,t} + (1 - \lambda_{i,1,t})\hat{\mu}_1\right] - \left[\lambda_{i,0,t}x_{i,0,t} + (1 - \lambda_{i,0,t})\hat{\mu}_0\right]}{\sqrt{\left(\sigma_{i,1,t}^{-2} + \hat{\tau}_1^{-2}\right)^{-1} + \left(\sigma_{i,0,t}^{-2} + \hat{\tau}_0^{-2}\right)^{-1}}}\right)$$

where $\Phi(\cdot)$ is the standard Normal CDF.

Using the principles of Thompson Sampling, at the next time point $t+1$, we will choose action 1 with probability $P(\theta_{i,1} > \theta_{i,0})$ and action 0 with $1 - P(\theta_{i,1} > \theta_{i,0}) = P(\theta_{i,0} > \theta_{i,1})$.

$$\mathcal{A}_i^{(t+1)} = \begin{cases} 1, & \text{with probability } P(\theta_{i,1} > \theta_{i,0}), \\ 0, & \text{with probability } 1 - P(\theta_{i,1} > \theta_{i,0}). \end{cases}$$

4. Based on the actions selected by the action selection strategy, the environment will then generate another reward, the learning algorithm will generate another set of estimates, and the action selection strategy will repeat this procedure.